

NexusRep

Clean Requirements Specification

Source-Agnostic, Vendor-Agnostic, CRM-Compatible AI Engagement Platform

Updated with data linkage, controlled agent flow, latency-aware judges, metrics rationale, and CRM handling

Document purpose

This document consolidates the NexusRep product requirements into a cleaner, non-duplicative specification. It makes explicit how inputs are linked, how data is stored and retrieved, how the controlled agent operates, how latency is managed, what metrics are required, and how CRM systems or market analogues handle those metrics.

The specification uses third-person requirement language. The MVP is deliberately scoped as one vertical slice: one brand, one therapeutic area, one approved content set, one AI rep experience, one DocNexus HCP target list, audit logging, and CRM-ready export.

Prepared based on the DocNexus NexusRep project specification, the prior requirements draft, and updated product discussion.

1. Executive Summary and Product Boundary

NexusRep shall be a modular AI engagement platform for pharma and medtech whitespace. It shall use DocNexus claims intelligence to identify high-opportunity HCPs, brand-approved content to control what may be said, a controlled agent graph to conduct the conversation, and CRM-compatible events to hand results back to the brand.

The product shall not be positioned as an avatar vendor, generic AI sales rep, or CRM replacement. It shall be positioned as the pharma-specific orchestration layer that links targeting, approved content, compliant conversation, auditability, analytics, and CRM handoff.

Boundary	Requirement
Buy/adapt	Avatar rendering, ASR, TTS, video transport, and CRM systems shall be treated as replaceable providers.
Build	DocNexus targeting, content normalization, controlled retrieval, conversation orchestration, compliance guardrails, audit, analytics, and CRM-compatible event mapping shall be built as NexusRep core.
MVP rule	The MVP shall prove the vertical slice before live CRM integration or enterprise omnichannel deployment.
Non-negotiable	No response shall be spoken or displayed unless it is backed by approved content and passes the final compliance gate.

2. Core Architecture: Interchangeability Without Rewrites

Every external source and vendor shall pass through an adapter. Core services shall only depend on canonical NexusRep objects, never on raw CRM payloads, PPT structures, PDF text, Tavus objects, ElevenLabs responses, or vendor-specific API formats.

External source or vendor

- > Adapter
- > Normalizer / mapper
- > Canonical NexusRep object
- > Core service logic
- > Output adapter / vendor adapter / CRM adapter

Area	Interchangeability requirement
Input sources	DocNexus claims, CRM export, scripts, PPT, PDF, DOCX, approved FAQ, approved website, CSV, and manual upload shall normalize into canonical objects.
AI vendors	Whisper, ElevenLabs, Mascot SDK, Tavus, HeyGen, D-ID, or text-only fallback shall be swappable through common interfaces.
CRM outputs	Veeva, Salesforce, IQVIA, Microsoft Dynamics, or CSV/JSON export shall be added through output adapters and field mapping.
Storage	The source of truth shall remain canonical storage; retrieval indexes and vendor caches shall not become product truth.

3. Clean Target Architecture

Frontend apps

- Brand Studio
- HCP AI Rep Experience
- Analytics and Compliance Console

Backend core

- Input Adapter Layer
- Canonical Data Layer
- Content and MLR Service
- Claims Targeting Service
- Retrieval Service
- Conversation Orchestrator
- Risk Classifier and Policy Router
- Response Builder and Validator
- Compliance Gate
- Escalation and Follow-Up Service
- Audit Service
- CRM Event Service
- Analytics Service
- Vendor Adapter Layer

Storage

- PostgreSQL as source of truth
- Object storage for original files/media

- Vector index for approved-content retrieval
- Append-only audit/event log
- Knowledge graph later for complex relationships

The first build may be a modular monolith with strong internal boundaries. The module boundaries shall be designed so that services can be split later without changing business logic.

4. Data Linkage, Storage, and Retrieval Architecture

The product shall use a hybrid storage model. PostgreSQL shall govern canonical truth. Object storage shall hold original files and recordings. A vector index shall retrieve relevant approved content. A knowledge graph may be added later for complex relationships. The audit log shall prove what happened.

Layer	Stores	Reason
PostgreSQL	Tenants, brands, campaigns, HCP profiles, opportunity scores, content metadata, approved answers, MLR metadata, sessions, turns, CRM events.	It provides transactions, versioning, foreign keys, approval states, tenant isolation, and the canonical source of truth.
Object storage	Original PPT/PDF/DOCX/script files, rendered slides, transcripts, audio/video recordings, consent files, exported reports.	Large files and immutable artifacts should not live inside relational tables.
Vector index / pgvector	Embeddings for approved answers, approved claims, FAQ entries, refusal templates, and slide text.	It supports semantic retrieval but never decides whether content is allowed.
Audit/event log	Append-only events for classification, retrieval, validation, compliance decisions, response output, escalation, and CRM export.	It proves traceability and supports compliance review.
Knowledge graph - future	Relationships among HCPs, indications, brands, claims, content, CRM ownership, territories, and escalation paths.	It helps when relationship logic becomes too complex for simple joins.

The retrieval rule shall be: RAG retrieves candidates; PostgreSQL validates eligibility; the response validator checks the answer; the compliance gate approves or blocks; the audit log proves the decision.

4.1 Entity linkage model

```
Tenant
-> Brand
  -> Campaign
    -> ApprovedContentSet
      -> ContentAsset
      -> ApprovedAnswer / ApprovedClaim / SafetyStatement / DetailAidSlide
    -> HCPProfile
      -> HCPOpportunityScore
    -> ConversationSession
      -> ConversationTurn
        -> IntentRiskClassification
        -> RetrievalEvent
        -> SourceValidation
        -> ResponseValidation
        -> ComplianceDecision
        -> ResponseOutput
        -> EscalationEvent
        -> AuditRecord
        -> CRMEvent
```

All modules shall link through canonical IDs such as tenant_id, brand_id, campaign_id, hcp_id, content_asset_id, approved_answer_id, safety_statement_id, session_id, turn_id, audit_event_id, and crm_event_id.

5. Ingestion and Normalization Flow

Raw source data shall never directly control a promotional or medical response. Each source shall be parsed by an adapter, normalized into canonical objects, reviewed or tagged with MLR metadata, embedded if eligible, and only then made available to retrieval.

```
PPT / PDF / script / CRM export / DocNexus list
-> source adapter
-> parser and normalizer
-> canonical object creation
-> MLR metadata validation
-> active approved content store
-> embedding generation
-> vector index
-> retrieval service
```

Input	Canonical objects created	Consistency control
Brand PPT/PDF/script	ContentAsset, ApprovedAnswer, ApprovedClaim, SafetyStatement, DetailAidSlide.	Requires MLR status, approval ID, expiry date, audience, market, and version.

DocNexus claims/HCP data	HCPProfile, HCPopportunityScore, TargetList, PersonalizationBrief.	Only HCP-level aggregate summaries enter the agent context; raw patient-level claims stay out.
CRM export	HCPProfile enrichment, territory owner, prior touchpoints, opt-out/consent, CRM owner mapping.	CRM is optional enrichment; the product must run without live CRM.
Approved website/knowledge base	ContentAsset and approved answer blocks if the brand authorizes the source.	Unapproved web content is quarantined and not used for live responses.

6. Controlled Retrieval and Consistency Rules

Retrieval shall be compliance-aware. The vector search shall only return candidate object IDs. The system shall resolve each candidate back to canonical records and validate content status before the response builder can use it.

Doctor question
-> semantic retrieval over approved embeddings
-> candidate approved_answer_id / slide_id / safety_statement_id
-> PostgreSQL source validation
-> response construction from approved blocks
-> response validation
-> final compliance gate

Consistency mechanism	Requirement
Canonical IDs	Every derived object shall retain links to source file, source version, MLR approval, campaign, and tenant.
Versioning	Approved content shall never be overwritten in place. New MLR-approved text creates a new version; prior versions remain audit-visible.
Approval state machine	Only ACTIVE + APPROVED + NOT_EXPIRED content may be retrieved for live responses.
Vector freshness	Embeddings shall store object_version and active flag. Retired content embeddings shall be inactive for future retrieval but retained for audit traceability.
Outbox pattern	CRM exports shall be written to a CRM outbox table first, then delivered asynchronously with retry and status tracking.

7. Controlled Agent Architecture

The NexusRep agent shall be a controlled graph, not a single free-form chatbot. The LLM may assist with classification and response assembly, but it shall not be the authority for promotional, efficacy, dosing, comparative, or safety claims.

HCP input
-> Input capture / ASR
-> Combined intent and risk classifier
-> Policy router
-> Approved answer path
-> Off-label refusal path
-> AE / pharmacovigilance path
-> Medical information / MSL path
-> Human follow-up path
-> Safe fallback path
-> Retrieval and source validation
-> Response builder
-> Response validator
-> Final compliance gate
-> Avatar / TTS / detail aid output
-> Audit + metrics + CRM event

Agent component	Role
Conversation Orchestrator	Controls the graph execution, loads session context, calls services, routes the turn, and creates output events.
Combined Intent/Risk Classifier	Returns intent, confidence, off-label risk, AE risk, medical-information risk, prompt-injection risk, and ISI requirement in one optimized call.
Policy Router	Routes the turn to approved answer, refusal, escalation, sample, human handoff, or fallback path.
Retrieval Service	Finds candidate approved content using metadata-filtered vector retrieval.
Source Validator	Checks MLR approval, expiry, audience, geography, indication, campaign, and content version.
Response Builder	Composes the response from approved blocks and controlled transitions; it does not create new medical claims.
Response Validator	Checks groundedness, unsupported claims, safety wording, ISI, PHI exposure, and route correctness.
Final Compliance Gate	Approves or blocks the exact response before it is spoken or displayed.
Audit and CRM Event Writer	Records the turn, evidence, decision, output, and CRM-ready event.

7.1 Latency-aware judge design

Judges shall be logical controls, not necessarily separate runtime LLM calls. The system shall judge everything, but it shall not LLM-call everything. The runtime shall combine checks, parallelize operations, and move non-critical review to asynchronous QA.

Risk	Latency-aware requirement
Too many LLM calls	Intent, off-label, AE, medical-info, and prompt-injection detection shall be combined into one classifier where possible.
Sequential processing	Retrieval, deterministic metadata checks, and risk classification should run in parallel when safe.
Slow validation	Most validation shall be deterministic: approved status, expiry, audience, campaign, exact ISI text, source IDs, and block IDs.
Overloaded runtime	Deep LLM audit, quality scoring, content-gap clustering, and long session summaries shall run after the turn or after the session.
Unsafe speed optimization	If latency conflicts with compliance correctness, the system shall fail safe and prioritize compliance.

Low-latency critical path:

ASR -> combined risk classifier + retrieval in parallel -> source validation
-> approved-block response -> fast validator -> compliance gate -> avatar output

Asynchronous path:

deep QA review -> analytics aggregation -> CRM retry/export -> content gap analysis

8. End-to-End Information Flow

The product shall separate setup-time information flow, runtime turn flow, and post-session flow.

Phase	Information flow	System output
Setup-time	Brand uploads approved content; DocNexus provides HCP target list; optional CRM export enriches owner, territory, history, and opt-out data.	Campaign, approved content set, target list, HCP briefs, vendor config, CRM mapping.
Runtime turn	HCP asks question; agent classifies risk; retrieves approved content; validates source and response; compliance gate approves; avatar speaks and shows detail aid.	Safe response, slide display, compliance decision, audit event.
Post-session	Session is summarized; metrics update; escalation/follow-up tasks are created; CRM-ready event is exported or queued.	Audit package, analytics row, CRM event, follow-up queue, content gap signals.

9. CRM Strategy and Market Alignment

NexusRep shall be CRM-compatible first and CRM-integrated later. The MVP shall export clean CSV/JSON activity records. Future releases may add Veeva, Salesforce, IQVIA, or custom CRM adapters. CRM systems shall be treated as systems of record for HCP/account relationships, activities, tasks, emails, consent, and field follow-up; they shall not become the agent brain.

Market system	What it already handles	NexusRep integration posture
Veeva Engage / Vault CRM	Digital HCP engagement, video meetings, compliant chat, approved content sharing, and activity capture inside CRM.	NexusRep should write AI detail activity, content shown, follow-up task, and compliance flags into CRM-compatible records.
Veeva Approved Email	Approved templates, fragments, optional personalization, and compliant rep-sent email tracking.	NexusRep should select approved follow-up templates or create CRM tasks, not generate free-form pharma emails.
Doceree RepTwin	Virtual pharma brand rep across video, voice, text, EHRs, journals, email, chat, and websites, trained on MLR-approved content according to public material.	NexusRep should differentiate through DocNexus claims-ranked targeting, transparent source validation, auditability, and CRM-ready outcome events.
Salesforce Life Sciences Cloud	AI/data-driven omnichannel HCP engagement and unified life sciences data/CRM workflows.	NexusRep should map events to Salesforce activity/task/campaign objects when a brand uses Salesforce.
IQVIA Field Force Agent / OCE-style workflows	Insights, segmentation, profiling, and compliant HCP engagement support for field teams.	NexusRep should complement next-best-action workflows by becoming a compliant AI engagement execution channel.

9.1 Canonical CRM event

```
{
  "event_type": "AI_DETAIL_COMPLETED",
  "hcp_npi": "1234567890",
```

```

"brand_id": "brand_001",
"campaign_id": "campaign_001",
"session_id": "session_001",
"duration_seconds": 420,
"completed_detail": true,
"content_shown": ["slide_1", "slide_3", "isi_1"],
"topics_discussed": ["dosing", "safety"],
"followup_requested": true,
"followup_type": "human_rep",
"off_label_flag": false,
"ae_flag": false,
"isi_delivered": true,
"summary": "HCP completed the AI detail and requested human follow-up."
}

```

10. Metrics Framework: What to Capture, Why It Matters, and CRM Handling

Metrics shall be designed for three audiences: commercial teams need reach and ROI, compliance teams need proof, and CRM/field teams need actionable follow-up. Each metric shall have a clear reason, data source, and CRM handling strategy.

10.1 Targeting and whitespace metrics

Metric	Why it is required	Where it comes from	CRM / market handling
Eligible patient density	Ranks HCPs by relevant patient opportunity, especially where human reps are not covering the account.	DocNexus claims-derived aggregate HCP features.	Usually not stored directly in CRM unless the brand maps it to target score/segmentation fields; Doceree-style platforms use clinical intent signals for activation.
Decile and growth trend	Identifies 1-6 decile HCPs that may be low current value but high future opportunity.	DocNexus claims/Rx/market data or brand segmentation imports.	CRMs often store decile/target tier as account attributes; NexusRep should export opportunity segment, not raw claims.
Whitespace category	Separates no-rep, under-covered, and no-see HCPs to justify AI engagement.	DocNexus + territory/CRM coverage data.	CRM may store owner/territory/access restrictions; NexusRep should map this to campaign targeting and activity rationale.
Recommended approved topic	Connects the HCP opportunity to an approved conversation starting point.	Opportunity score + approved content metadata.	CRM can receive topic of interest or next-best-action style recommendation.

10.2 Engagement metrics

Metric	Why it is required	Where it comes from	CRM / market handling
Incremental HCP reach	Shows how many HCPs were reached beyond the field force or existing digital channels.	Target list, session starts, CRM history if available.	Veeva/Salesforce-style CRMs can store activity records; Doceree-style platforms report reach across channels.
Outreach-to-session rate	Measures whether invitations or entry points drive doctors into the AI rep experience.	Invite logs, landing events, session starts.	CRM campaigns can track invite and response; Approved Email-style tools track sends/opens/clicks when used.
Session completion rate	Shows whether the HCP stayed long enough to receive the core message.	ConversationSession status and duration.	CRM activity can store completed AI detail as a call/activity outcome.
Average conversation duration	Indicates depth of engagement and helps compare digital rep economics with rep calls.	Conversation timer.	CRMs commonly store call duration or activity details; NexusRep should export duration_seconds.
Human follow-up requests	Strong intent signal and direct handoff to field/medical teams.	HCP action button or conversation intent.	CRM task/event should be created for assigned rep, MSL, or medical information team.

10.3 Content and conversation metrics

Metric	Why it is required	Where it comes from	CRM / market handling
Topics discussed	Shows what HCPs care about and guides future content planning.	Intent classifier and approved answer IDs.	CRM may receive a concise topic summary; full transcript should not be pushed unless configured.
Content shown	Proves which approved detail aids, ISI, or resources were displayed.	DetailAidSlide IDs and response output events.	Veeva-style systems are content-centric; NexusRep should map slide/content IDs to CRM content records when possible.
Content gap count	Identifies common HCP questions with no approved answer, which informs MLR/content teams.	Fallback events and no-match retrieval logs.	Usually kept in NexusRep analytics; may become content request/task for brand teams.
Approved answer usage	Shows which MLR-approved blocks are actually useful in conversations.	ApprovedAnswer IDs in retrieval and audit records.	Can be exported as content performance analytics rather than per-HCP CRM data.

10.4 Compliance and safety metrics

Metric	Why it is required	Where it comes from	CRM / market handling
ISI delivery rate	Must prove safety information was delivered when required.	Compliance gate and output events.	Audit console is primary; CRM can store isi_delivered=true for the activity.
Off-label refusal rate	Shows how often the agent correctly avoided prohibited promotion.	Risk classifier, policy router, refusal template usage.	CRM should usually get an escalation flag and task, not sensitive full wording.
AE capture and routing rate	Required for pharmacovigilance workflow and safety accountability.	AE classifier, AE capture node, escalation service.	CRM may only store safe AE flag/status; full AE details should route to PV systems according to policy.
Unapproved response count	Zero-tolerance metric proving the system did not speak unsupported claims.	Response validator and compliance gate.	Should remain internal compliance KPI; only summary status should be shared externally.
Sessions fully auditable	Shows each interaction has transcript, source IDs, MLR IDs, decisions, and output logs.	Audit event completeness checks.	Compliance console is primary; CRM may store audit package link.

10.5 ROI and operational metrics

Metric	Why it is required	Where it comes from	CRM / market handling
Cost per completed detail	Core economic comparison against human rep coverage and static digital campaigns.	Platform cost + completed sessions.	Usually NexusRep analytics; CRM can provide baseline field activity volume.
Cost per follow-up request	Measures efficiency of creating actionable sales/medical opportunities.	Cost data + follow-up task count.	CRM task outcomes can close the loop when reps/MSLs act.
Rx/claims lift - future	Ultimate commercial outcome metric, but not MVP-critical because it needs time and controls.	Claims/Rx outcome data and matched controls.	Doceree-style platforms discuss measuring prescription/fill/refill; NexusRep should add outcome analytics later.
CRM export success rate	Ensures activity records reach the brand system reliably.	CRM outbox status.	Veeva/Salesforce/IQVIA adapters should expose sent, failed, retry, and mapped-field errors.
Runtime latency	Controls HCP experience quality while preserving compliance.	ASR, classifier, retrieval, validation, TTS/avatar timing.	Usually internal operational metric; severe latency may affect engagement completion.

11. Compliance, Security, and Audit Requirements

Control	Requirement
Approved-content-only	The response builder shall use approved blocks, source IDs, and controlled transitions. It shall not invent product claims.
Final compliance gate	Every exact response shall be approved or blocked before avatar/TTS output.
Fail-safe behavior	When classification, retrieval, or validation is uncertain, the system shall refuse, escalate, or use an approved fallback.
PHI minimization	Raw patient-level claims shall not enter live conversations or third-party AI/avatar/TTS/ASR vendors in the MVP.
Tenant isolation	Brand data, HCP lists, content, audit logs, CRM mappings, and exports shall be isolated per tenant.
Immutable audit	Audit records shall be append-only in production; corrections shall be appended as correction events, not destructive edits.

12. Current MVP Scope

Included in MVP	Explicitly excluded from MVP
One brand, one therapeutic area, one approved content set, one AI rep or brand persona.	Live Veeva/Salesforce/IQVIA API integration.
Manual/PPT/PDF/script ingestion and normalization.	Full CRM bidirectional sync.
DocNexus HCP target list and HCP-level opportunity score import.	Open web scraping or unapproved web knowledge.
Whisper + ElevenLabs + Mascot SDK prototype stack with Tavus adapter design.	Raw patient-level claims in live AI conversation.
Controlled retrieval, response validation, final compliance gate, audit log.	Automated free-form pharma email generation.
Basic analytics, CRM-ready CSV/JSON export, CRM outbox pattern.	Rx lift/matched-control outcome proof in first MVP.

13. Future Scope

Future area	Capabilities
Input adapters	Live Veeva, Salesforce Life Sciences Cloud, IQVIA, Veeva Vault/PromoMats, approved website, SharePoint/Box/Drive, EHR/media channel sources.
Vendor adapters	Tavus production CVI, HeyGen, D-ID, Synthesia, custom WebRTC avatar, text-only assistant.
Output adapters	Veeva activity write-back, Salesforce task/activity creation, IQVIA call log write-back, Approved Email trigger, PV system routing, BI export.
Advanced analytics	Rx lift, claims outcome tracking, matched controls, decile movement, market-share lift, competitive switching, territory optimization.
Enterprise compliance	Full MLR workflow, legal hold, retention controls, region-specific policy, human QA queue, formal PV integration.
Knowledge graph	Dynamic relationships among HCPs, claims-derived segments, content eligibility, CRM ownership, territories, and escalation paths.

14. Testing and Deployment Requirements

Test type	Requirement
Unit tests	Adapters, normalizers, content versioning, retrieval filters, CRM mapping, risk classifier, source validator, response validator, compliance gate.
Integration tests	Content upload to approved retrieval; HCP question to safe response; off-label refusal; AE escalation; CRM event creation; audit completeness.
Red-team tests	Off-label prompts, competitor comparisons, safety paraphrase attempts, prompt injection, missing content, AE language, PHI leakage attempts.
Vendor swap tests	Same conversation shall work through Whisper + ElevenLabs + Mascot SDK, Tavus, and text-only fallback without changing core behavior.
Latency tests	Measure ASR, classifier, retrieval, validation, compliance gate, TTS/avatar start, and total perceived response time.
Source swap tests	Approved answers from PPT, PDF, script, CRM export, and knowledge base shall normalize into the same canonical model.

A successful compliance test run shall show zero unapproved claims, zero unsafe off-label answers, complete source traceability, correct AE escalation, correct ISI delivery when required, and no raw claims/PHI sent to third-party vendors.

15. Source Notes and Market References

This document is based on the DocNexus NexusRep project specification, the previous NexusRep requirements draft, and the user-provided markdown notes. Public market references used for CRM/market handling include the following official or product sources:

- Veeva Vault CRM Engage product page and resources: digital HCP meetings, compliant chat, approved content sharing, and CRM capture.
- Veeva Approved Email product/help pages: approved templates, fragments, optional personalization, and interaction tracking through Veeva CRM.
- Doceree RepTwin product page and blog: virtual pharma brand representative across channels, trained on MLR-approved content, with HCP question/engagement analytics.
- Salesforce Life Sciences Cloud product pages: AI/data-driven omnichannel HCP engagement and life-sciences engagement workflows.
- IQVIA Field Force Agent and omnichannel engagement materials: HCP segmentation, dynamic targeting, compliant engagement planning, and field-team insights.

Final clean requirement statement

NexusRep shall support any approved source in, one canonical internal data model, one controlled latency-aware agent graph, one compliance gate before output, and any CRM or AI vendor out. The product shall be architected so that adding a new input source, CRM, or avatar vendor requires only a new adapter and mapping, not changes to targeting, compliance, retrieval, audit, analytics, or core conversation logic.